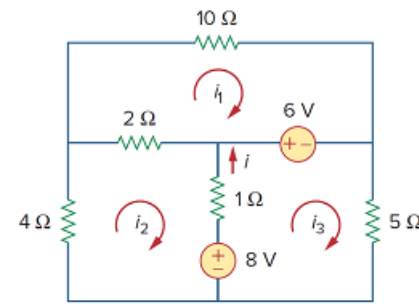


Problem

Apply mesh analysis to find i in Fig. 3.87.

Figure 3.87:



Step-by-step solution

Step 1 of 3

Refer to Figure 3.87 in the textbook.

Apply mesh analysis to loop1.

$$10i_1 - 6 + 2(i_1 - i_2) = 0$$

$$12i_1 - 2i_2 = 6 \quad \text{..... (1)}$$

Apply mesh analysis to loop2.

$$4i_2 + 2(i_2 - i_1) + 1(i_2 - i_3) + 8 = 0$$

$$-2i_1 + 7i_2 - i_3 = -8 \quad \text{..... (2)}$$

Step 2 of 3

Apply mesh analysis to loop3.

$$6 + 5i_3 - 8 + 1(i_3 - i_2) = 0$$

$$-i_2 + 6i_3 = 2 \quad \text{..... (3)}$$

Solve equations (1), (2) and (3).

$$i_1 = 0.329 \text{ A}$$

$$i_2 = -1.0256 \text{ A}$$

$$i_3 = 0.1624 \text{ A}$$

Step 3 of 3

Calculate the value of current, i .

$$i = i_3 - i_2$$

Substitute 0.1624 A for i_3 and -1.0256 A for i_2 .

$$\begin{aligned} i &= 0.1624 - (-1.0256) \\ &= 1.188 \text{ A} \end{aligned}$$

Thus, the value of current, i is 1.188 A .

